

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — TOPICAL WORKSHEET PACK: 15C.2

PREPARATIONS AND REACTIONS OF CARBOXYLIC ACIDS

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	Topic 15: ORGANIC CHEMISTRY, CARBONYLS AND CHIRALITY
SUB-TOPIC:	15C.2: Preparations and Reactions of Carboxylic Acids
TOTAL QUESTIONS:	3 Questions (Structured + MCQs + Calculations)
TOTAL MARKS:	8 Marks
EXAMINER FAQs:	1 Empirical Examiner FAQs & Traps Included

HOW TO USE THIS TOPICAL PACK:

1. Complete all structured questions under timed exam conditions.
2. Pay close attention to command words (Explain, Deduce, Calculate, State).
3. Review the complete worked mark scheme in Section C.
4. Study the Empirical Examiner FAQs in Section D to avoid common candidate traps.

SECTION A: TOPICAL PRACTICE QUESTIONS — 15C.2

Q1. 1. (a) Write equations for the synthesis of propanoic acid starting from (i) propan-1-ol and (ii) propanenitrile.

[4 Marks]

WCH14/01/Jun22/Q8(a)

Q2. (b) Describe the test to distinguish a carboxylic acid from an alcohol using sodium hydrogencarbonate, NaHCO_3 .

[2 Marks]

WCH14/01/Jun22/Q8(b)

Q3. (c) Write the reaction equation for ethanoic acid with PCl_5 and state the observation.

[2 Marks]

WCH14/01/Jun22/Q8(c)

SECTION B: WORKED MARK SCHEME — 15C.2

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: 1. (a) Write equations for the synthesis of propanoic acid starting from (i) propan-1-ol and (ii) propanenitrile. 1. (i) $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH} + 2[\text{O}] \rightarrow \text{CH}_3\text{CH}_2\text{COOH} + \text{H}_2\text{O}$ (Reagents: $\text{K}_2\text{Cr}_2\text{O}_7 / \text{H}_2\text{SO}_4$, reflux). 2. (ii) $\text{CH}_3\text{CH}_2\text{CN} + \text{HCl} + 2\text{H}_2\text{O} \rightarrow \text{CH}_3\text{CH}_2\text{COOH} + \text{NH}_4\text{Cl}$ (Reagents: dilute HCl, reflux).	[4]
Q2	Q2: (b) Describe the test to distinguish a carboxylic acid from an alcohol using sodium hydrogencarbonate, NaHCO_3 . 1. Add NaHCO_3 (aq). 2. Carboxylic acid produces effervescence / CO_2 gas (turns limewater cloudy); alcohol shows no reaction.	[2]
Q3	Q3: (c) Write the reaction equation for ethanoic acid with PCl_5 and state the observation. 1. $\text{CH}_3\text{COOH} + \text{PCl}_5 \rightarrow \text{CH}_3\text{COCl} + \text{POCl}_3 + \text{HCl}$. 2. Steamy / misty white fumes of HCl gas observed.	[2]

SECTION C: EMPIRICAL EXAMINER FAQS & COMMON TRAPS — 15C.2**FAQ #1: Why does NaHCO_3 react with carboxylic acids but not phenols or alcohols?****Category: Acidity
Comparison****COMMON EXAMINER TRAP / MISTAKE:**

Thinking phenols are strong enough acids to liberate CO_2 from carbonates.

MODEL ANSWER / EXAMINER EXPECTATION:

Carboxylic acids are stronger acids ($K_a \sim 10^{-5}$) than carbonic acid, so they displace CO_2 from NaHCO_3 . Phenols ($K_a \sim 10^{-10}$) and alcohols are weaker acids than carbonic acid, so no reaction occurs.